

Program : Diploma in Printing Technology	
Course Code : 6107	Course Title: Press Standardization Lab
Semester : 6	Credits: 2.5
Course Category: Program Core	
Periods per week: 4 (L:0 T:1 P: 3)	Periods per semester: 60

Course Objectives:

- To impart the knowledge of procedure of implementing standardization process in Press
- To assess the effects of standardization process on the quality of print.
- To provide hand on experience on quality checking of raw materials & printed output, setting of different units of machine according to standards and Printing on substrate maintaining consistency.

Course Prerequisites:

Topic	Course code	Course name	Semester
Printing on Offset machine.		Advanced Printing Machines Workshop I NSQF	3
Offset printing mechanism.		Mechanism of printing machines-II	3
Measurement of quality parameters of printed product.		Fundamentals of Color Reproduction	3
Print Quality measurements.		Color Analysis Lab	3
Imposition, Layout preparation, Platemaking		Digital Imaging Techniques	4
Plate making Techniques.		Master Preparation Workshop	4

Course Outcomes:

On completion of the course, the student will be able to:

COn	Description	Duration (Hours)	Cognitive Level
CO1	Preparation of Test forms and Plate Output	18	Applying

CO2	Dampening solution preparation and Machine test & Print the test form	11	Applying
CO3	Assessment of Test form	12	Applying
CO4	Remedial measures for standardizing the press	15	Applying
	Lab Exam	4	

CO – PO Mapping:

Course Outcomes	PO1	PO2	PO3	PO4	PO5	PO6	PO7
CO1	3						
CO2	3					3	
CO 3	3	3		3		3	
CO4	3		3				

3-Strongly mapped, 2-Moderately mapped, 1-Weakly mapped

Course Outline:

Module Outcomes	Description	Duration (Hours)	Cognitive Level
CO1	Preparation of Test forms and Plate Output.		
M1.01	Choose a standard Testform.	8	Applying
M1.02	Prepare layout.	2	Applying
M1.03	Create print ready file.	2	Applying
M1.04	Ripping & Plate output.	3	Applying
M1.05	Check for plate linearity.	3	Applying
CO2	Dampening solution preparation and Machine test & Print the test form.		
M2.01	Adjust Press room conditions.	1	Applying
M2.02	Prepare dampening solution-Check the pH, Conductivity, Temperature & hardness.	2	Applying
M2.03	Machine setting–Check blankets. Set the Nip pressure settings. Roller pressure setting	2	Applying

M2.04	Setting the inking unit.	2	Applying
M2.05	Printing-Print the test form.	4	Applying
	Lab Exam- I	2	Applying
CO3	Evaluation of Test form.		
M3.01	Measure the Lab values of the patches, calculate ΔE .	6	Applying
M3.02	Measure the dot gain, contrast and gray balance.	4	Applying
M3.03	Output the curve and evaluate with standards.	2	Applying
CO4	Remedial measures for standardizing the press.		
M4.01	Adjust the quality parameter of the printed chart to the standard value by resetting the machine.	15	Applying
	Lab Exam- II	2	

Text /Reference:

T/R	Book Title/Author
R1	HL Apfelberg and MJ Apfelberg, Implementing quality management in graphic arts ,GATF
R2	P. Kipphan, Handbook of Print Media , Springer, 2002.
R3	Bob thomson Printing Material Science and Technology PIRA
R4	Michael Bernard, John Peacock. Handbook of printing and production
R5	Heigh. M. Speir, Introduction in Printing Technology .
R6	Adams J.M, Printing Technology 5th edition , Delmar Publishers, NewYork.
R6	Prameela J Groff , Test images for printing .

Online Resources:

Sl.No	Website Link
1	https://www.rsgsolutions.co.in/solution/new-press-standardization/
2	<a href="https://www.heidelberg.com/global/media/en/global_media/products___prinec
t_modules/pdf/color_and_quality/standardization_in_offset_printing.pdf">https://www.heidelberg.com/global/media/en/global_media/products___prinec t_modules/pdf/color_and_quality/standardization_in_offset_printing.pdf